Identifying Congested and Uncongested Regions in the Barcelona Road Network

Arthus Duval, Nicolas Künzli

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Abstract

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2.1 The SimBarca dataset

The dataset is stored in plk files in which we can retrieve two important data types. The first being *vdist*, which stores the total travelled distance among every vehicles on every single link for every time intervals. The second is *vtime* which stores the total time spent on every links for every time intervals.
There are 101 different files, each of them representing a different simulation called session. They are each in a folder labelled as such: *session_###* where the three last characters are digits representing the session id. The folder *session_000* contain the simulation with a demand rate of 100%, the following 20 are with a demand rate of 120%, the following with 140%, and so on with *session_081 - session_100* with a demand rate of 200%.
...

2.2 Network representation

The network is built around three main components: links, nodes and polygons. The first being the most important in our congestion analysis, we will focus our attention on it by presenting its main characteristics. The csv file contains these following columns: *id, from_x, from_y, to_x, to_y, length, out_ang, num_lanes*. The first column is the id of the link. The following four are the coordinates of the box containing the link in which two opposite corners are the starting and ending coordinates of the link. These corners are set by *out_ang* which is the angle orientation of the segment. The *length* column is the length of the segment, and finally, *num_lanes* are the number of lanes contained in the segment.
Note that the network is simplified by grouping each individual lane with its neighbouring lanes, regardless of their directions. Nodes represent the center of the links and are used for the connectivity matrix. Links that are connecting the same roads are grouped under the same node.
The blue polygons delineate the intersection areas. They are displayed solely for visualization purposes and are not involved in the network computations.

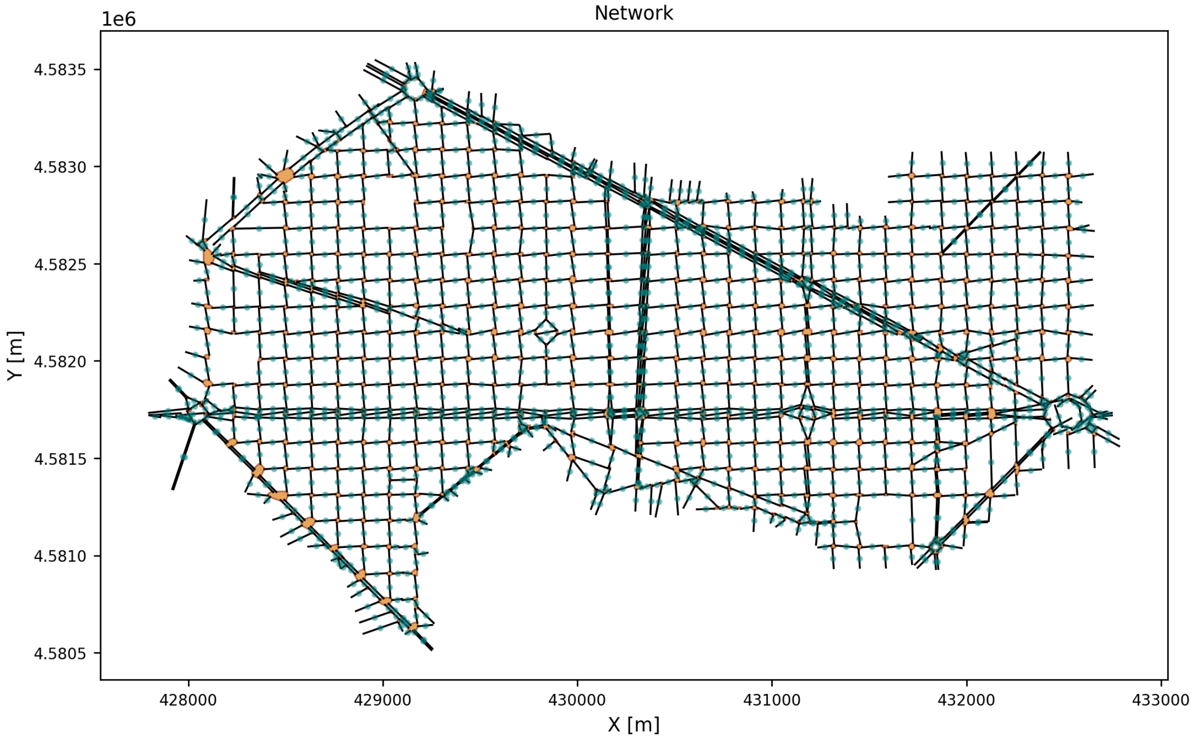


Figure 1: Simplified representation of the SimBarca road network. Links are represented as black segments, nodes as red points, and intersections as blue polygons.

As shown in Figure **??**, the network is represented as a graph whose nodes correspond to groups of connected links.

2.3 Temporal structure: sessions and 3-minute aggregation

As introduced in Section **??**, the dataset contains 101 simulation sessions grouped according to different demand rates. In the remainder of this work, the temporal evolution of traffic variables is analyzed using 3-minute aggregation intervals, allowing direct comparisons across sessions and demand scenarios. Although data were also available at a 5-second resolution, the 3-minute aggregation level was deemed sufficient for the objectives of this project while reducing short-term fluctuations and computational complexity.
Let $x_{s,t,l}$ denote the value of a traffic variable measured on link l during time interval t in session s. The session-averaged value is computed as

$$\bar{x}_{t,l} = \frac{1}{N_{t,l}} \sum_{s=1}^{S} x_{s,t,l}, \tag{1}$$

where S is the number of sessions included in the averaging process and $N_{t,l}$ is the number of valid (non-missing) observations available for link l and time interval t. Depending on the analysis, S may correspond either to all 101 sessions or to the subset of sessions associated with a specific demand level. Missing values are ignored during the averaging process. If no valid observation exists for a given pair (t, l), the corresponding average is left undefined (`NaN`).
For the SimBarca dataset, each demand level above the baseline is represented by 20 independent simulation sessions, while the 100% demand level is represented by a single session. The resulting session-averaged matrices preserve the original spatial and temporal dimensions of the data while reducing the variability associated with individual simulation runs.

2.4 Speed reconstruction

The SimBArCa dataset provides two complementary speed signals for each road segment ij at each 3-minute timestep t.

Aggregated trajectory speed (`pred_speed`). Vehicle-trajectory aggregates supply two cumulative fields per segment per bin: $\mathrm{vdist}_{ij}(t)$ [m], the total distance covered by all vehicles crossing segment ij in the 3-minute window, and $\mathrm{vtime}_{ij}(t)$ [s], their total travel time. Space-mean speed is then:

$$v_{ij}^{\mathrm{pred}}(t) \;=\; \frac{\mathrm{vdist}_{ij}(t)}{\mathrm{vtime}_{ij}(t)}. \tag{2}$$

When $\mathrm{vtime}_{ij}(t) = 0$ (no vehicle observed), the ratio is undefined and stored as NaN; the physical interpretation and imputation policy are developed in Section **??**.

Loop-detector speed (`ld_speed`). A second signal is delivered directly by the simulated loop-detector model as a pre-computed speed [m/s]. It is generated from the same underlying trajectories but aggregated through a different sensor model, producing a slightly smoother time-series that is better suited to free-flow normalisation (see Section **??**).

Both signals share the same temporal grid (3-minute bins, 08:03–09:57), the same spatial coverage ($N = 1\,570$ directed segments), and the same upper bound imposed by the simulation's universal $50\,\mathrm{km\,h^{-1}}$ speed limit ($\approx 13.89\,\mathrm{m/s}$). The two uses in the analysis are:

- `pred_speed` (Equation **??**): used for the congestion-index characterisation below and as the speed input to the Grid K-means clustering (Section **??**).
- `ld_speed`: used for percolation analysis (Section **??**), where both the numerator $v_{ij}(t)$ and the free-flow denominator $v_{\max,ij}$ are drawn from the same sensor to avoid cross-sensor bias.

Congestion index and network-wide dynamics. As a first characterisation of the speed field, a *congestion index*

$$\mathrm{CI}_{ij}(t) \;=\; \min\left(1,\; \max\left(0,\; 1 - \frac{v_{ij}^{\mathrm{pred}}(t)}{v_{ij}^{\mathrm{ff}}}\right)\right), \tag{3}$$

is defined for each segment, where v_{ij}^{ff} is the 85th-percentile of `pred_speed` for segment ij pooled over all sessions and timesteps — a robust proxy for free-flow conditions. A segment with $\mathrm{CI} > 0.40$ operates below $60\,\%$ of its free-flow speed, a standard threshold for moderate-to-severe congestion.

Figure **??** shows the *network-wide congestion share* — the fraction of valid segments exceeding $\mathrm{CI} > 0.40$ — as a function of time, averaged over all sessions. Two regimes stand out.

1. **Slow early build-up (08:03–09:15):** the share rises gradually from about $8\,\%$ to $18\,\%$, reflecting localised queuing on the principal arterials while the broader network remains near free-flow. The flat plateau around 08:30–09:00 corresponds to the sustained congestion phase identified in Section **??**.
2. **Late-peak acceleration (09:15–09:57):** from roughly 09:15 the share climbs more steeply, reaching approximately $40\,\%$ by 09:57. This second phase captures queue spill-back from saturated junctions propagating outward to secondary roads.

Critically, the congestion share remains *below 50 %* throughout the entire analysis window (dashed reference line), confirming that even at the height of the rush hour more than half the road segments are still moving at adequate speed. This is consistent with the percolation threshold $q_c = 0.52$ (Section **??**): the network's giant functional cluster only begins to fragment at that threshold, not before.

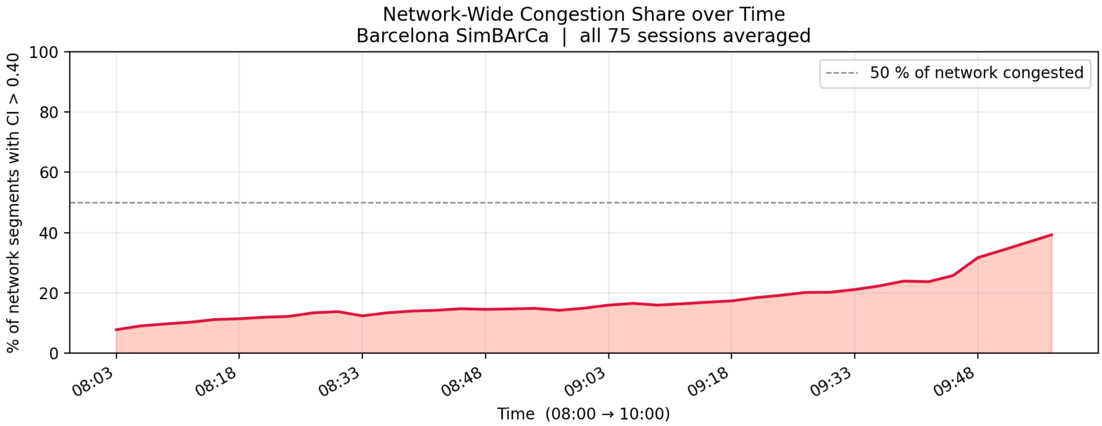


Figure 2: Network-wide congestion share over time (08:03–09:57; $T = 39$ steps of 3 minutes each, averaged over all sessions). The y-axis gives the fraction of road segments with congestion index CI > 0.40, i.e. operating below 60 % of their free-flow speed. The dashed line marks the 50 % reference. The two-phase pattern — slow early build-up followed by steeper late-peak acceleration — is consistent with localised queue formation giving way to network-wide spillback.

2.5 Missing-data analysis

2.5.1 Spatial and temporal distribution of NaNs

A missing value (NaN) in the speed matrix arises when the aggregate vehicle travel time $\mathrm{vtime}_{ij}(t) = 0$: no vehicle traversed segment ij during the three-minute interval ending at time t. In that case the ratio $v_{ij}(t) = \mathrm{vdist}_{ij}(t)/\mathrm{vtime}_{ij}(t)$ is arithmetically undefined and is stored as NaN rather than zero. This distinction is physically essential: a zero speed would incorrectly label the segment as fully congested (stopped traffic), whereas a NaN correctly signals the *absence of observations*. Conflating the two—as an earlier version of the codebase did by calling `np.nan_to_num(speed, nan=0.0)`— artificially inflates congestion metrics and biases any downstream clustering or percolation analysis toward finding congestion where none exists.

Spatial pattern. Figure **??** (left panel) shows the sensor coverage map coloured by the per-segment NaN fraction, averaged across all 101 sessions and all 39 timesteps of the analysis window. Two spatial regimes are visible.

- *Major corridors* (primary arterials, ring roads, and the principal routes connecting the main origin–destination zones): near-zero NaN fractions. During the morning peak these segments are continuously occupied by queuing or moving traffic, so every 3-minute bin contains at least one vehicle.

- *Peripheral and access links* (short stubs, intersection-internal links, exclusive ramps, low-demand residential streets): persistently high NaN fractions. These segments receive few vehicles even at peak hour because the simulated OD demand matrix routes most trips along the principal arterials. The right panel of Figure **??** highlights the segments whose NaN fraction exceeds 50 % of all timesteps.

A small subset of segments have no valid speed observation anywhere in the dataset; they are treated as permanently excluded from all analyses.

Temporal pattern. Missing data are not uniformly distributed over the 39 timesteps. In the very first intervals (08:03–08:15), when morning congestion is still developing and traffic volumes on secondary roads are lower, NaN fractions are marginally elevated. As the peak intensifies (08:30–09:15) the main corridors saturate and their NaN fraction approaches zero, while lightly used segments remain sparse throughout. Figure **??** shows the distribution of NaN fractions pooled over all $101 \times 1\,570$ segment–session pairs, revealing a strongly bimodal pattern: a sharp mass near zero (well-observed arterials at peak hour) and a broader tail at high NaN fractions (structurally low-demand links).

Spatial autocorrelation. To determine whether missing data cluster geographically—a key question for the viability of spatial interpolation—the Pearson correlation between each segment's NaN fraction and the mean NaN fraction of its direct road-network neighbours was computed (Figure **??**). A correlation $r \gg 0$ indicates that entire sub-areas of the network are simultaneously under-sampled (an area-level demand deficit), whereas $r \approx 0$ implies idiosyncratic, link-level gaps. The observed correlation directly informs the NaN handling policy below.

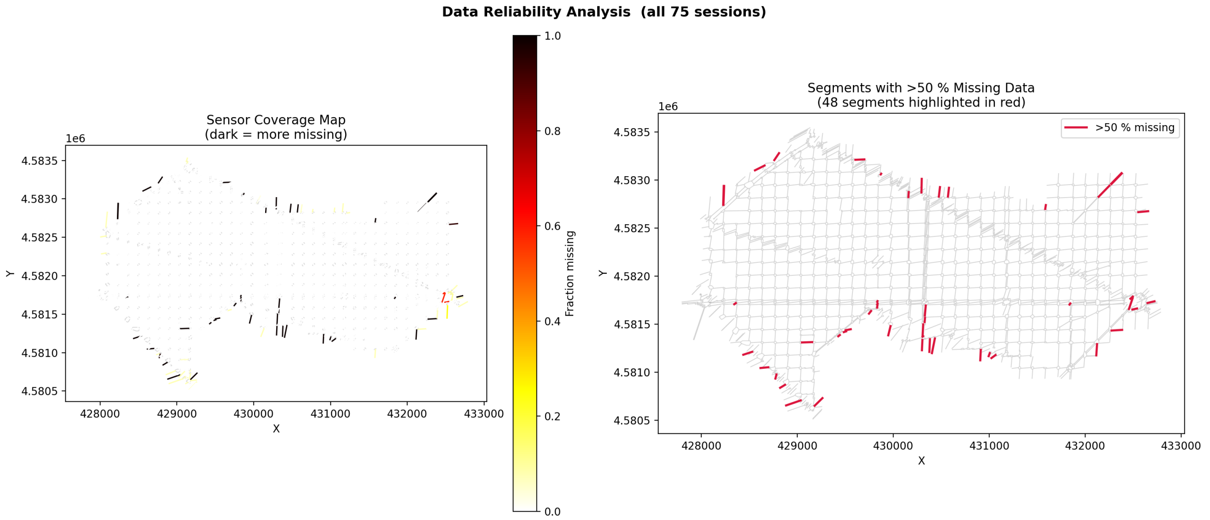
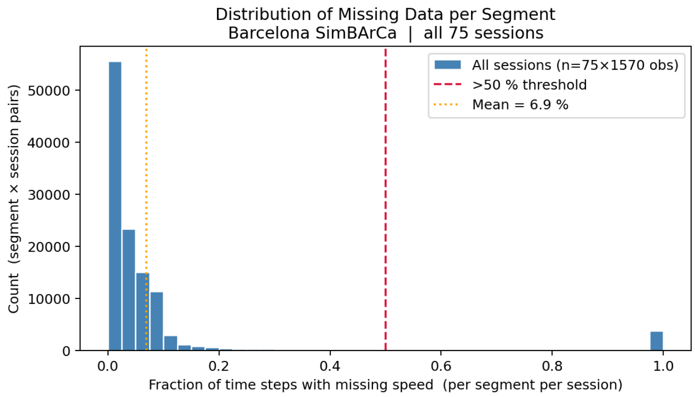
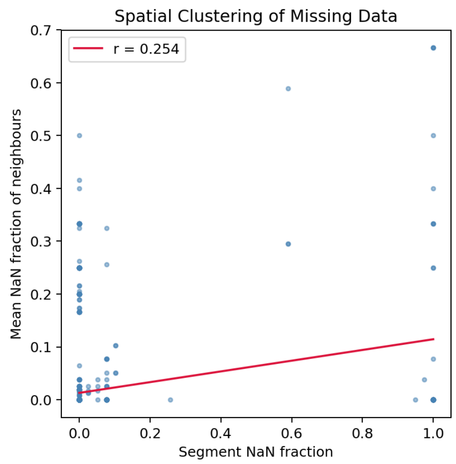


Figure 3: Sensor coverage of the SimBArCa network (all 101 sessions, 08:03–09:57). **Left**: NaN fraction per segment averaged over sessions and time (dark = more missing; `hot_r` colourmap from 0 to 1). **Right**: segments whose time-averaged NaN fraction exceeds 50 % of all timesteps, highlighted in crimson against the light-grey network.



(a) Distribution of NaN fractions across all 101 × 1 570 segment–session pairs. The dashed red vertical line marks the 50 % visualisation threshold; the dotted orange line is the overall mean NaN rate. Most pairs cluster near zero (well-observed corridors); a right tail captures persistently sparse links.



(b) Spatial autocorrelation of NaN fractions. Each point is a segment; the x-axis is its own NaN fraction and the y-axis is the mean NaN fraction of its road-network neighbours. The Pearson r (shown on the regression line) quantifies geographic clustering of missing data.

Figure 4: Missing-data diagnostics for the SimBArCa speed matrix.

2.5.2 NaN handling policy

A principled imputation strategy must satisfy two competing requirements: it must avoid injecting spurious congestion signals (never replace NaN with zero), and it must not discard so many segments that the spatial structure of the network becomes fragmented for analysis. After evaluating three candidates, the following tiered policy was adopted and implemented in `processing/speed.py::fill_speed_nans()`.

Why not zero-fill? Replacing NaN with speed $= 0$ is physically incorrect. A segment with no observed vehicles during a 3-minute bin is simply unoccupied—most likely *free-flowing* at or near the speed limit during the brief gap—not blocked at standstill. Zero-filling would create dense phantom congestion clusters in areas that are actually clear, corrupting both the percolation threshold detection and the clustering analysis. This approach was rejected outright.

Case A — Temporal-median fill ($\leq$ 30% NaN). If fewer than 30 % of a segment's $T = 39$ timesteps are missing (at most 11 out of 39), each missing value is replaced by the segment's own *temporal median* computed from the non-missing timesteps. Three properties justify this choice.

1. *Representativeness*: with at least 28 valid observations the median is a stable estimate of the segment's typical speed during the rush-hour window.
2. *Robustness to outliers*: the median is insensitive to isolated spikes from brief queues or sensor noise; the mean would be pulled by such extremes, potentially filling NaNs with unrealistically high or low values.
3. *Segment specificity*: each segment is filled with its own typical speed, preserving spatial heterogeneity. A fast arterial's fill value is high; a slow approach link's is low. This is especially important for percolation, where the normalised ratio $r_{ij} = v_{ij}/v_{\max,ij}$ must reflect the true congestion state of each individual link.

After filling, Case A segments participate normally in all downstream computations (clustering, percolation, descriptive statistics).

Case C — No-data exclusion (> 30% NaN). Any segment exceeding the 30 % threshold is flagged in a Boolean `nodata_mask` and rendered in neutral grey (`#cccccc`) on all maps; it contributes neither to summary statistics nor to clustering or percolation features. The 30 % cut-off is motivated as follows.

- Below 30 % NaN the temporal median is well-anchored (at least 28 data points); above it, the available observations may be dominated by the pre-congestion early period, giving a misleadingly high fill value that would make a chronically slow segment look faster than it is.
- Segments above the cut-off are structurally low-demand links for which any imputation is speculative. Honest exclusion prevents the analysis from being distorted by fabricated speed values.
- In practice the proportion of excluded segments is small enough that the principal network topology is preserved for percolation and clustering.

Case B — Spatial neighbour fill (discarded). A third candidate—replacing NaN values with a weighted mean of adjacent segments' speeds via the road-network adjacency matrix—was explored but rejected. In a directed urban network, graph-adjacent segments often operate in radically different speed regimes: a residential stub feeding directly onto a major arterial, or an on-ramp abutting a free-flow motorway, will have a speed profile entirely unlike its topological neighbours. Averaging across such heterogeneous adjacent links injects substantial bias rather than recovering meaningful signal. The rejection is captured by the guiding principle: *an honest grey is preferable to a misleading imputed value.* Case B is therefore merged into Case C: any segment that cannot be reliably filled from its own temporal history is excluded from the analysis rather than interpolated spatially.

Implementation note. `fill_speed_nans()` is always called *after* `mean_over_sessions()`, which collapses the session dimension via `nansum/valid_counts` to preserve NaN-awareness. All visualisation functions check `nodata_mask` before assigning a colour to a segment; flagged segments receive `NODATA_COLOR`. A second threshold parameter `nodata_threshold=0.90` is retained in the function signature as a reserved hook for a future, non-adjacency-based imputation strategy (e.g. corridor-mate fill via geometric grouping), but is currently inactive.

2.6 Network-level descriptive statistics

2.6.1 Average speed map

Figure **??** shows the *average speed map*: for each road segment the colour encodes its mean speed averaged over all 101 simulation sessions and all 39 timesteps of the morning-peak analysis window. The RdYlGn colour scale runs from dark red (near standstill) through yellow to dark green (near free-flow, i.e. close to the simulated 50 km/h speed limit, $\approx 13.9\,\text{m/s}$).

Three spatial regimes emerge:

- *Congestion hotspots* (dark red): persistent on the arterial corridors that channel the dominant commuting flows toward the city centre. These segments are congested for a significant fraction of the 08:03–09:57 window across all demand realisations, suggesting structural bottlenecks independent of random session-to-session variation.

- *Free-flow periphery* (green): segments at the outer edges of the study area, and roads not on principal commuting routes, remain at or near the speed limit throughout the peak period. They are rarely congested because the simulated OD demand matrix routes most traffic away from them.
- *Intermediate links* (orange-yellow): corridors that are free-flowing during the early peak (08:03–08:20) but queue up at the height of the rush hour (08:30–09:15), yielding intermediate time-averaged speeds. These links are of special interest for the percolation analysis: they sit close to the network's tipping point and their congestion state is the most sensitive to demand fluctuations.

Because the map averages over both sessions and time, it collapses the dynamic evolution of congestion into a static *congestion fingerprint* of the network. It is useful for identifying the most structurally problematic links but does not reveal when or how quickly congestion builds; the temporal dimension is addressed in Section **??**.

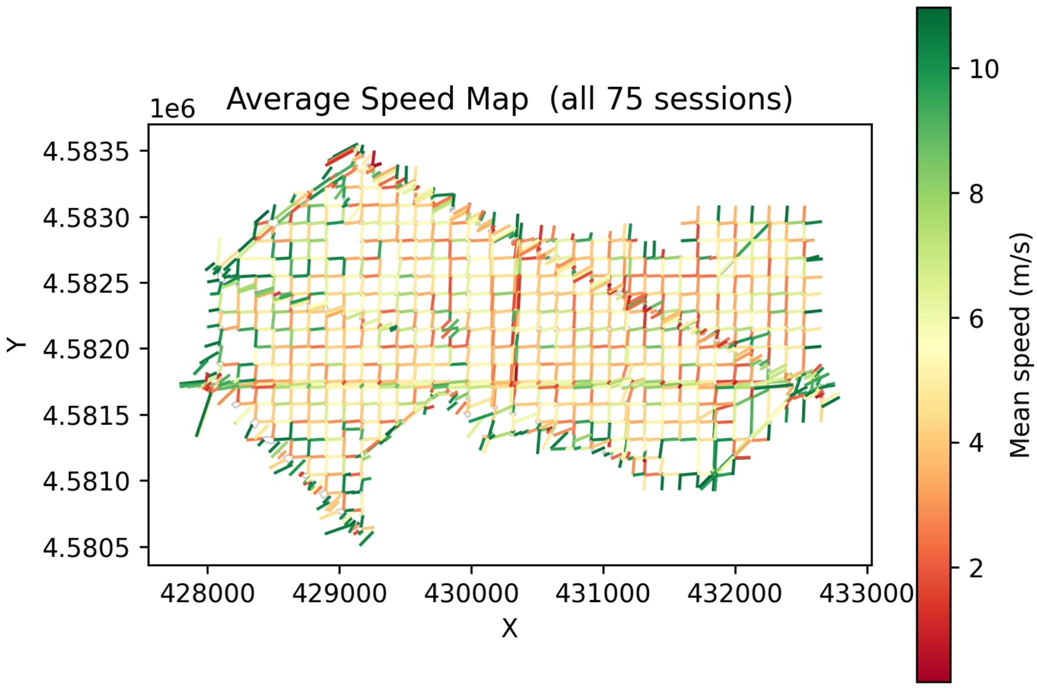


Figure 5: Average speed map of the SimBArCa road network (all 101 sessions, 08:03–09:57). Each segment is coloured by its mean speed across all sessions and timesteps (RdYlGn; green = fast, red = slow). Grey segments have more than 30 % missing observations and are excluded from all statistics (Case C, Section **??**).

2.6.2 Network mean speed over time

Figure **??** shows the evolution of the *network-mean speed*—the spatial mean over all valid (non-grey) segments—across the 08:03–09:57 window. Each of the 101 individual session traces is drawn in light blue; the thick line is the cross-session mean.

The curve exhibits the canonical signature of a simulated morning rush hour in three phases:

1. **Rapid onset** (08:03–∼08:30): the network-mean speed drops sharply as the first wave of commuters saturates the principal arterials. The steep descent indicates that the network transitions from near free-flow to congested conditions within the first 30 minutes of the analysis window.

2. **Sustained congestion plateau** (∼08:30–09:15): speed stabilises at its lowest level. The network is in a near-equilibrium between vehicles entering and leaving the study area; the broad arterials operate well below free-flow speed, and secondary roads receive spillback from saturated junctions. The width of this plateau in the session-mean curve—and the relatively compact spread of individual traces—confirms that the congestion regime is stable and reproducible across demand realisations.
3. **Gradual recovery** (∼09:15–09:57): as simulated demand decreases, queue lengths shorten and the network-mean speed rises slowly back toward free-flow conditions. The recovery is noticeably asymmetric (slower than the onset) because traffic must redistribute across the entire network before speeds normalise; this hysteresis is a well-known property of urban traffic dynamics.

The spread among the 101 session traces quantifies the session-to-session variability driven by different random demand realisations in the Simbarca simulator. The relatively compact envelope around the cross-session mean validates the use of session-averaged quantities as representative inputs for the clustering and percolation analyses in Section **??**.

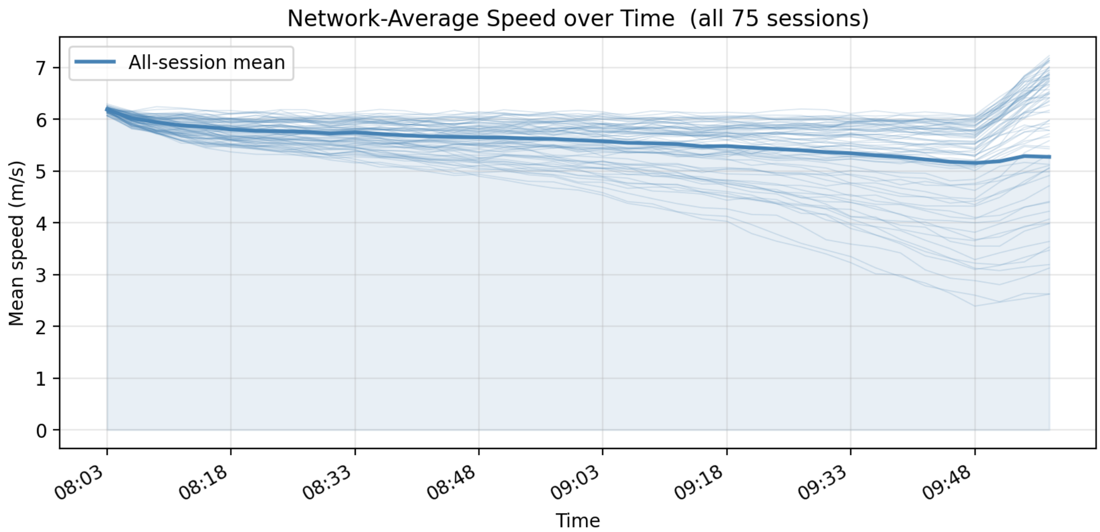


Figure 6: Network-average speed over time (08:03–09:57; $T = 39$ steps of 3 minutes each). Light blue lines: individual session traces (101 sessions). Thick blue line: cross-session mean with shaded area. The rapid drop, plateau, and slow recovery are the characteristic phases of the simulated morning rush hour.

2.6.3 Slowest segments

Figure **??** highlights the 20 road segments with the lowest time-and-session-averaged speed. These are the *chronically congested links* of the simulated Barcelona network: roads that remain near standstill for a disproportionate fraction of the morning peak, independently of the specific demand realisation. Because the ranking aggregates over all 101 sessions and all 39 timesteps, a segment that appears in this set is slow not just in high-demand outlier sessions but *systematically* across all runs of the simulator.

Two spatial properties stand out:

- **Spatial clustering**: the 20 slowest segments are not scattered randomly across the network but form tight spatial groups, typically located at or near major junctions, merging sections, and narrow approach links feeding high-capacity arterials. This is consistent with the network topology funnelling large vehicle volumes through a small number of physical constrictions.

- **Candidate bottlenecks**: segments that are chronically slow in the descriptive sense are strong candidates for the network bottlenecks identified by the percolation framework in Section **??**. The percolation approach provides a rigorous, topology-aware confirmation of whether these links are indeed the *critical bridges* whose congestion causes the giant functional cluster to fragment.

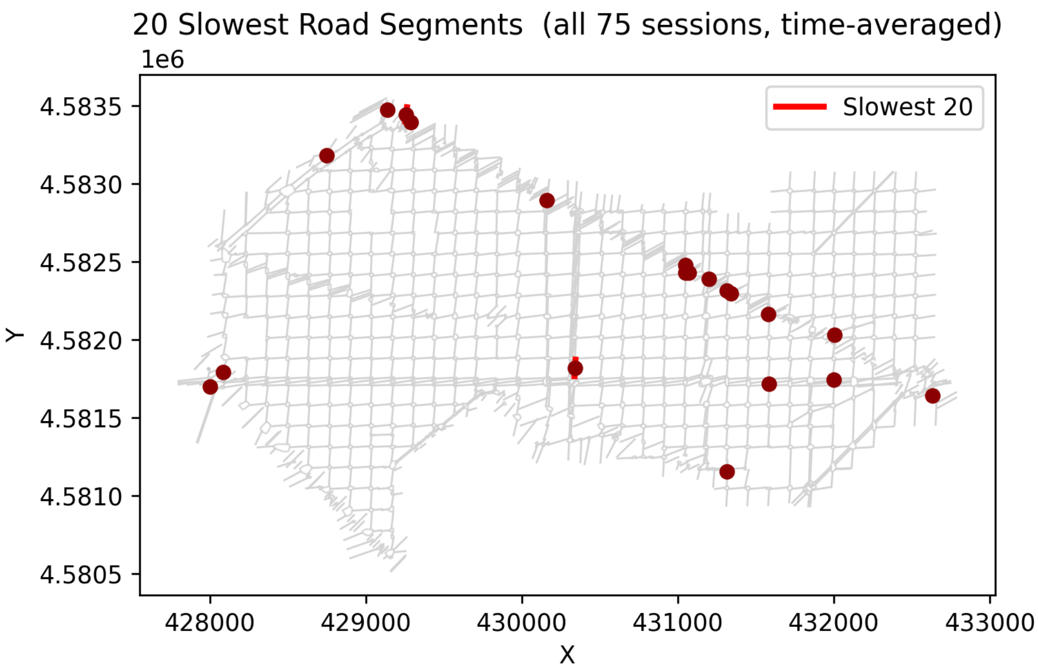


Figure 7: The 20 road segments with the lowest time-and-session-averaged speed (highlighted in red with centroid dot), superimposed on the full network (light grey). Segments are ranked by ascending mean speed over all 101 sessions and 39 timesteps. Their spatial clustering around a small number of junctions reveals the structural bottlenecks of the network.

2.7 Network simplification

The raw SimBArCa directed graph contains $N = 1\,570$ road segments. A significant proportion of these correspond to the same physical road: a two-way street is encoded as two antiparallel directed arcs; a multi-lane carriageway may appear as several parallel arcs of slightly different length. Left unsimplified, this duplication has two adverse effects.

- **Visual.** Overlapping arcs drawn in different colours occlude each other on speed and percolation maps, making the spatial pattern difficult to interpret.
- **Statistical.** Computing group-level statistics (e.g. the per-group mean speed or the location of bottleneck segments) on the raw graph gives undue weight to roads with many parallel representations, skewing both the clustering feature vectors and the descriptive results.

Grouping algorithm

The simplification is implemented in `network/simplify.py::group_segments()` using three geometric matching criteria; any single criterion is sufficient to merge two segments i and j into the same group:

1. **Forward match.** Both endpoint pairs are within $d_{\text{ep}} = 35\,\text{m}$: $\|p_i^{\text{from}} - p_j^{\text{from}}\| \le d_{\text{ep}}$ *and* $\|p_i^{\text{to}} - p_j^{\text{to}}\| \le d_{\text{ep}}$. This captures same-direction parallel lanes.

2. **Reverse match.** The endpoints cross-pair within d_{ep}: $\|p_i^{\text{from}} - p_j^{\text{to}}\| \le d_{\text{ep}}$ *and* $\|p_i^{\text{to}} - p_j^{\text{from}}\| \le d_{\text{ep}}$. This groups the two opposing-direction arcs of a two-way road.
3. **Lateral match.** Segment midpoints are within $d_{\text{lat}} = 15\,\text{m}$. This catches geometrically adjacent arcs of different lengths (e.g. a short access stub alongside a longer through-lane).

All matching pairs are fed into a path-compressed union–find (disjoint-set) forest, which merges them into equivalence classes in near-linear time. Within each group the *representative* segment is the member with the highest lane count, the natural proxy for road hierarchy.

Scope of application

The simplified network is used exclusively for **visualisation** and **descriptive statistics**:

- Percolation maps and bottleneck overlays (Section **??**) draw one arc per physical road, coloured by SCC membership or speed ratio.
- The average speed map (Section **??**) colours each representative by the nanmean speed of its entire group.

All topology-sensitive computations — SCC detection, the percolation sweep, and BFS cluster analysis — operate on the full 1 570-segment directed graph to preserve the exact connectivity structure of the road network. The simplification layer is a purely cosmetic and statistical convenience that does not alter the underlying analysis.

3 Identifying Congested and Uncongested Regions

3.1 Overview and Evaluation Framework

Three complementary approaches are used to identify congested and uncongested regions in the Barcelona road network during the morning peak period (08:03–09:57, $T = 39$ timesteps of three minutes each).

Although the three methods rely on different clustering strategies, they all aim to partition the network into regions exhibiting distinct traffic states. The comparison is performed on a common set of speed measurements and evaluation metrics.

Approach 1 (Section ??) — Threshold-based connected-component detection. A global congestion threshold q_c is applied to the road network. Road segments whose normalised speed falls below the threshold are classified as congested. The resulting binary congestion map is projected onto a spatial grid and Breadth-First Search (BFS) is used to identify connected congested regions.

Approach 2 (Section ??) — Link-based K-means clustering. Road segments are clustered directly using feature vectors combining spatial coordinates and temporal speed characteristics. This approach groups links with similar traffic behaviour while preserving geographical coherence through spatial weighting and manually selected centroid spawn points.

Approach 3 (Section ??) — Grid K-means clustering. The road network is first projected onto a 20×16 spatial grid covering the study area. Each grid cell is represented by its average normalised speed and spatial coordinates. K-means is then used to partition the grid into a fixed number of spatially coherent traffic regions.

Evaluation Metrics. The three approaches are compared using a common set of quantitative metrics. For each identified region k and timestep t, we evaluate:

- the median speed $\tilde{v}_k(t)$, which measures the internal traffic state and homogeneity of the region;
- the cluster size $S_k(t)$, defined as the number of links or grid cells belonging to the region;
- the number of congested regions $N_c(t)$, which characterises the spatial fragmentation of congestion across the network.

Together, these metrics provide a complementary assessment of traffic conditions, spatial extent, and congestion structure, allowing a direct comparison between the three approaches.

3.2 Approach 1 — Threshold-Based Grid Clustering

3.2.1 Spatial Grid Aggregation

Each of the 1 570 road segments is assigned to the grid cell containing its midpoint centroid. Segments with more than 30 % missing observations (Section **??**) are globally excluded and contribute to no cell. Segments that are unobserved at a specific timestep but are not globally excluded count as *uncongested* at that timestep: a NaN observation is not treated as evidence of congestion.

For the BFS clustering at each timestep t (Section **??**), the cell congestion state is determined by the fraction of present segments with $r_{ij}(t) < q_c$, as described above. For the static threshold-selection step (Section **??**), the mean normalised speed $\bar{r}_c$ (Equation **??**) is averaged over all sessions and all timesteps, and used to fit the K-means clustering of Approach 2 (Section **??**).

Trailing-NaN pre-fill before session averaging. Some simulation sessions end before 09:57, leaving trailing NaN values at the final timesteps of the analysis window. Without correction, the session-mean at late timesteps would be computed only over the subset of sessions that run to completion, introducing a selection bias if those sessions have atypical speeds. To prevent this, each session's trailing NaNs (those appearing *after* its last valid observation) are filled with the segment's free-flow reference value (1.0 for normalised r; the segment's 95th-percentile raw speed for m/s units) before the session mean is taken. This sets unobserved end-of-session timesteps as uncongested, which is the physically appropriate conservative assumption.

3.2.2 Congestion Threshold Selection

A key design choice is the value of the global threshold q_c. Two candidates are evaluated and compared.

Percolation-based threshold ($q_c = 0.52$, normalised speed). The threshold is derived objectively from the network's own traffic dynamics, following the percolation framework introduced by li2015percolation [**?**].

Intuition. The idea is borrowed from statistical physics, where *percolation* describes how a fluid spreads through a porous medium as more and more pores become open. Here, each road segment is the analogue of a pore: it is *open* (functional) if traffic moves fast enough, and *blocked* (congested) otherwise. As we lower the speed requirement for a road to be called functional, more roads qualify and the network of functional roads gradually fills up — much like water progressively filling a sponge. The critical question is: at what speed threshold does a *city-wide* connected path first emerge? That threshold is q_c.

Step 1 — Normalised speed ratio. For each road segment ij and each 3-minute timestep t, the speed ratio is:

$$r_{ij}(t) \;=\; \frac{v_{ij}(t)}{v_{\max,ij}}, \tag{4}$$

where $v_{ij}(t)$ is the loop-detector speed measured on segment ij at time t, and $v_{\max,ij}$ is the free-flow reference speed for that segment — taken as the maximum loop-detector speed observed on segment ij across all 101 simulation sessions and all recorded timestamps (pre-rush, rush and post-rush). Because the Simbarca simulation enforces a 50 km/h speed limit, $v_{\max,ij}$ converges to that limit ($\approx$ 13.9 m/s), giving $r_{ij} \in [0,\, 1]$ where $r = 1$ means free-flow and $r = 0$ means standstill.

Step 2 — Functional subgraph at threshold q. For a candidate threshold q, segment ij is declared *functional* if $r_{ij} \geq q$ and *dysfunctional* (congested) otherwise:

$$\text{segment } ij \text{ is functional} \iff r_{ij} \;\geq\; q. \tag{5}$$

Retaining only functional segments produces a *functional subgraph* of the road network. When $q = 0$ every road is functional (the full network); when $q = 1$ only roads at free-flow speed are functional (the network is nearly empty).

Step 3 — Strongly connected components (SCCs). The functional subgraph is a *directed* graph, because roads in the dataset are one-way: a road from A to B does not necessarily imply a road from B to A. A *strongly connected component* (SCC) is a maximal set of segments within which every segment is reachable from every other segment by following directed road edges — in other words, a group of roads where a driver can circulate freely without leaving the subgraph. The size of the *giant SCC* $G(q)$ is the number of segments in the largest such group; it measures how much of the network remains globally connected at threshold q.

Step 4 — Identifying the critical threshold q_c. As q increases from 0 to 1, the giant SCC shrinks because more roads become dysfunctional. Just before it finally breaks apart, a characteristic signature appears: the *second-largest SCC* $S_G(q)$ reaches a maximum. This peak marks the *percolation transition* — the precise moment when the giant cluster is about to fragment. The critical threshold is therefore defined as:

$$q_c \;=\; \underset{q \,\in\, [0,1]}{\arg\max}\; S_G(q). \tag{6}$$

The reasoning is as follows: just below q_c the network still forms one large connected cluster; just above q_c it has already broken into fragments. The second-largest cluster is largest precisely at the transition point, when the piece that is about to detach from the giant is at its maximum size before fully separating.

Step 5 — Computing q_c on the Barcelona network. In practice, q is swept in 100 steps from 0 to 1. At each step, the functional subgraph is constructed from the real road-adjacency matrix (1 570 directed segments), SCCs are computed, and $G(q)$ and $S_G(q)$ are recorded. The input speed field is the session-and-time-averaged normalised speed $\bar{r}_{ij}$ (mean over all 101 sessions and all 39 analysis-window timesteps). Figure **??** shows the resulting transition curve.

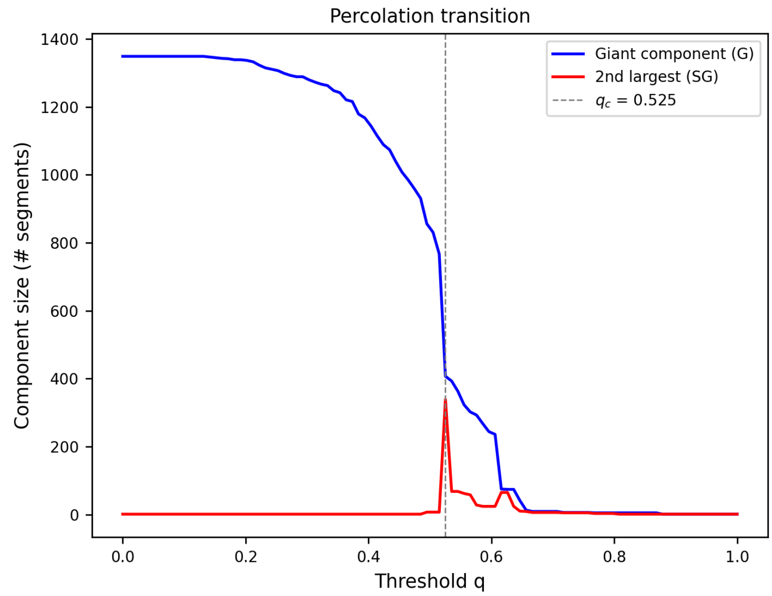


Figure 8: Percolation transition of the Barcelona road network. **Blue**: giant SCC size $G(q)$, i.e. the number of road segments forming the largest connected functional cluster as the threshold q increases. **Red**: second-largest SCC size $S_G(q)$. The dashed vertical line marks $q_c = 0.52$, the value at which S_G is maximised: above this threshold the giant cluster begins to fragment.

The peak of the red curve occurs at $q_c = 0.52$, meaning that the network undergoes a *connectivity transition* when the speed threshold is set at 52% of the free-flow speed. In absolute terms:

$$q_c \times v_{\max} = 0.52 \times 50\,\mathrm{km/h} \approx 26\,\mathrm{km/h}. \tag{7}$$

Roads travelling faster than 26 km/h are considered functional; roads below that speed act as barriers. At exactly $q_c = 0.52$, the network is balanced at its tipping point: the functional roads just barely form one large connected cluster. A slight increase in congestion (raising q above 0.52) causes that cluster to fragment into isolated sub-networks, trapping traffic in local pockets with no city-wide path. This provides a *physically grounded, data-driven* definition of network-scale congestion, in contrast to an arbitrary expert-chosen threshold.

The percolation threshold $q_c = 0.52$ is retained for the main analysis because it is data-driven, reproducible across sessions, and grounded in network topology. The raw-speed threshold ($q = 2.0\,\mathrm{m/s}$) is shown in parallel in Section **??** to illustrate the sensitivity to threshold choice.

3.2.3 Bottleneck Detection

As a by-product of the percolation sweep, segments whose normalised speed lies in the marginal band $r_{ij} \in [q_c - \delta,\ q_c + \delta]$ with $\delta = 0.01$ are identified as *bottlenecks*. These links are barely functional at q_c; their congestion would sever the bridges between functional sub-networks and collapse the giant SCC. Figure **??** shows their spatial distribution on the simplified network.

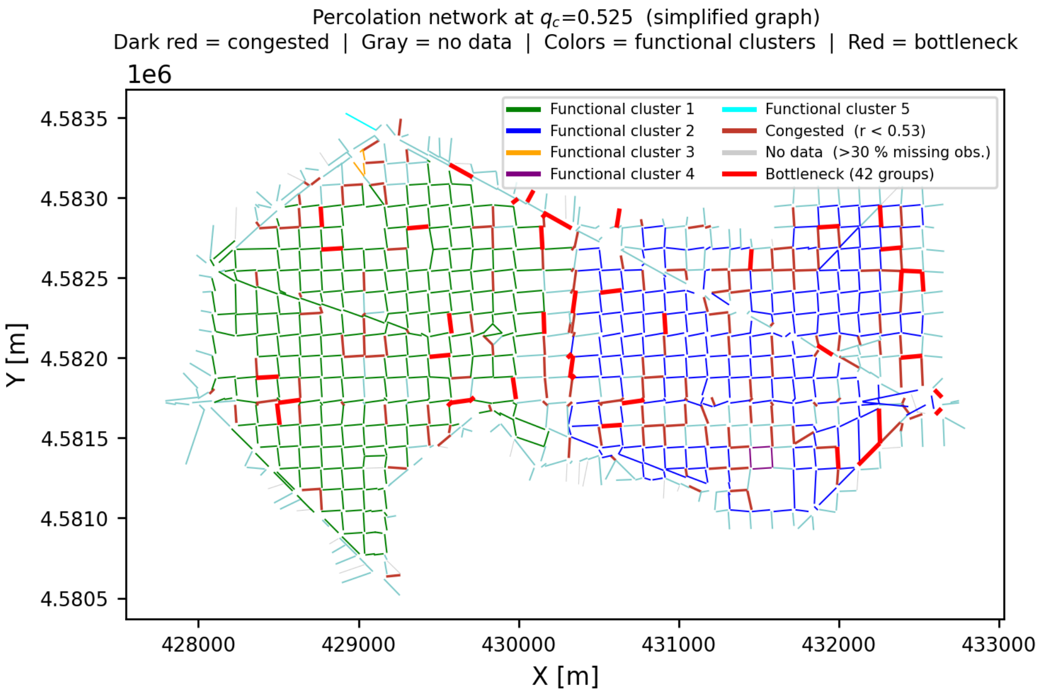


Figure 9: Network state at $q_c = 0.52$ (session-and-time-averaged, simplified graph). Colours denote strongly connected components (green = giant SCC; blue, orange, purple, cyan = smaller SCCs). Dark-red segments are congested ($r_{ij} < 0.52$); **thick red** segments are bottlenecks ($r_{ij} \in [0.51, 0.53]$); grey segments have insufficient data ($> 30\%$ missing observations).

3.2.4 BFS Connected-Component Analysis

At each timestep t, the binary congestion map is constructed by labelling every cell as *congested* if more than 40 % of its present road segments have $r_{ij}(t) < 0.52$. Segments that are unobserved at t but not globally excluded count as uncongested in this fraction, so missing observations do not artificially inflate the congestion signal. A BFS with **8-connectivity** (four cardinal directions plus four diagonals) then identifies every maximal spatially contiguous set of congested cells as one component. At the ≈ 250 m cell resolution, diagonal neighbours almost always share road infrastructure; restricting to four-connectivity would artificially split zones that are physically contiguous across a grid boundary. Components are ranked by descending size at each timestep (cluster 1 = largest congested zone). The set of all functional cells (congested fraction ≤ 40 %) forms a single *functional background* cluster; within it, the *largest functional connected component* is tracked separately as a proxy for network-wide connectivity.

3.2.5 Results: Congestion Maps and Temporal Metrics

Figure **??** shows the temporal evolution of the top-5 congested clusters under $q_c = 0.52$ (normalised speed, mean over all 101 sessions). The four-row layout presents: spatial cluster maps at four selected timesteps (row 1); congested cluster sizes over time, ranked by size at each timestep (row 2); median raw speed per congested cluster over time (row 3); and the size and median speed of the largest functional connected component over time (row 4).

Figure 10: Approach 1 — $q_c = 0.52$, normalised speed, mean over all sessions. **Row 1**: BFS cluster maps at four representative timesteps. Congested clusters are coloured by per-timestep size rank (cluster 1 = red, cluster 2 = blue, cluster 3 = orange, etc.); functional cells are green; no-data cells are grey. **Row 2**: number of grid cells in the top-5 congested clusters over time (ranked by size at each timestep independently). **Row 3**: median raw speed (m/s) of segments within each congested cluster over time. **Row 4**: size (grid cells, left axis) and median speed (m/s, right axis) of the largest functional connected component over time. Vertical dashed lines indicate the timesteps shown in row 1.

Figure ?? repeats the analysis with the manual raw-speed threshold $q = 2.0\,\text{m/s}$. This stricter criterion targets only near-standstill traffic and establishes an upper bound on how severe congestion must be to register as a connected zone under the manual threshold.

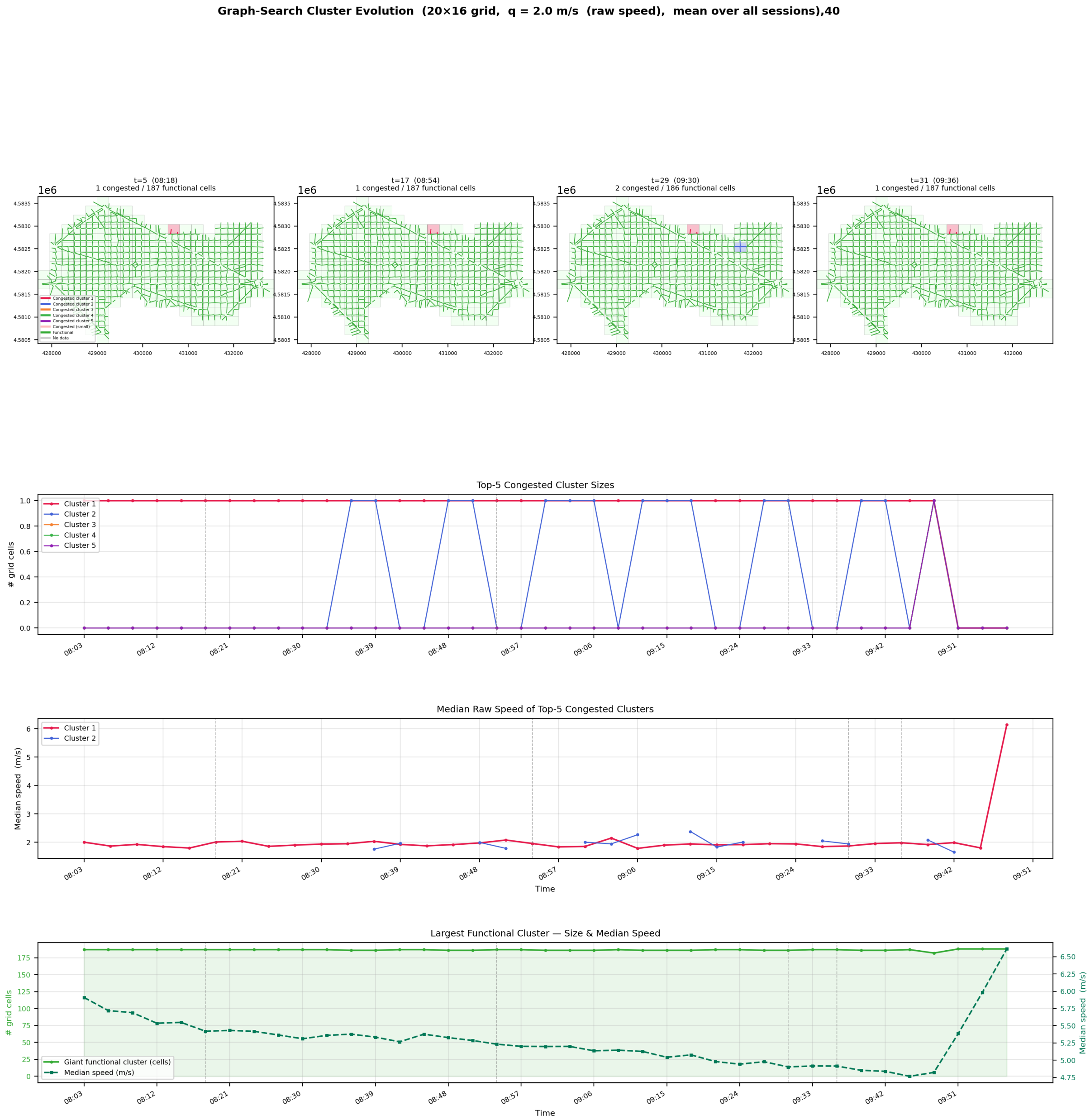


Figure 11: Approach 1 — $q = 2.0\,\mathrm{m/s}$, raw speed. Same four-row layout as Figure **??**. A cell is labelled congested if more than 40 % of its segments have raw speed $< 2.0\,\mathrm{m/s}$ ($= 7.2\,\mathrm{km/h}$), targeting only near-standstill traffic.

Table **??** reports the three evaluation metrics for Approach 1 at the 15 representative timesteps under $q_c = 0.52$. Clusters are ranked by size at each timestep; the functional background is reported separately.

Table 1: Approach 1 metrics ($q_c = 0.52$, normalised speed, mean over all sessions). N: number of congested BFS components. S_k: size (grid cells) of cluster k (ranked by size at that timestep). $\tilde{v}_k$: median raw speed (m/s) of that cluster. $\tilde{v}_{\text{bg}}$: median speed of the functional background.

Time	N	S_1	S_2	S_3	$\tilde{v}_1$	$\tilde{v}_2$	$\tilde{v}_3$	$\tilde{v}_{\text{bg}}$
08:03	0	–	–	–	–	–	–	5.90
08:12	2	1	1	–	3.44	2.86	–	5.57
08:18	3	2	1	1	3.31	3.05	3.19	5.46
08:24	5	2	2	1	3.55	2.81	2.54	5.48
08:30	4	2	2	1	3.85	3.34	2.73	5.36
08:42	5	2	1	1	3.16	2.66	2.85	5.33
08:54	4	2	2	1	3.48	2.93	4.58	5.28
09:00	4	2	2	1	2.99	2.59	3.79	5.26
09:06	5	2	2	2	3.49	3.46	2.66	5.21
09:12	6	2	2	2	3.37	3.81	3.01	5.20
09:18	7	2	2	2	3.51	3.73	2.37	5.14
09:30	6	4	3	3	4.01	3.66	2.69	5.03
09:42	8	6	5	4	4.08	3.85	3.96	5.02
09:51	10	19	13	4	3.80	3.83	3.50	5.19
09:57	10	41	8	3	3.87	4.12	4.23	5.64

3.3 Approach 2— K-means Clustering

3.3.1 Feature Construction and K-means Setup

Rather than applying a hard binary threshold, we use K-means to assign every link to a cluster based on features that jointly encode *spatial position* and *speed dynamics*. For each link c, the feature vector is built as

$$\mathbf{x}_c = \Big[w_s\, \tilde{x}^{(c)},\ w_s\, \tilde{y}^{(c)}; w_d\, \tilde{\mu}_c,\ w_d\, \tilde{\sigma}_c,\ w_d\, \tilde{t}_c^{\text{peak}},\ w_d\, \tilde{\mathbf{u}}_{c,1:8},\ \Big], \tag{8}$$

where μ_c and σ_c are the mean and standard deviation of the speed profile of link c, t_c^{peak} is the normalised time index of the minimum speed, and $\mathbf{u}_{c,1:8}$ are the first eight components extracted from the row-wise normalised temporal speed profile. The variables $x^{(c)}$ and $y^{(c)}$ denote the link-centre coordinates.

All dynamic and spatial features are standardised independently using `StandardScaler`, giving zero mean and unit variance before weighting. The parameters w_d and w_s control the relative importance of the dynamic speed features and the spatial coordinates, respectively. Thus, the final feature vector contains 11 speed-related features and 2 spatial features.

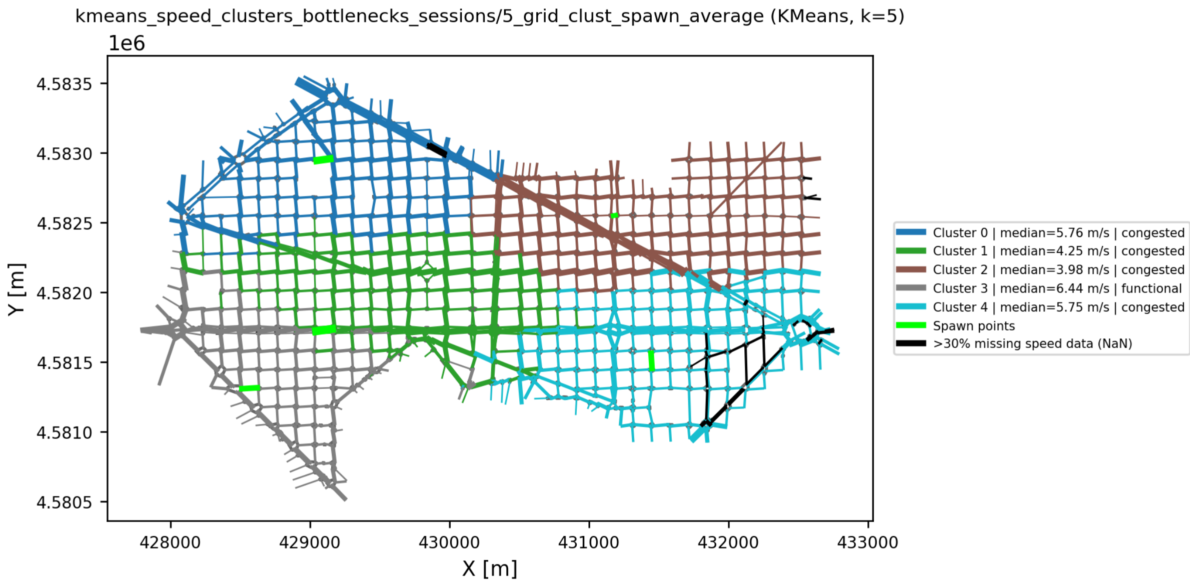
3.3.2 Cluster Count, Weight Parameters and Spawn points

Multiple numbers of clusters were evaluated, ranging from 5 to 9. A minimum of 5 clusters was considered necessary, as fewer clusters would merge distinct traffic patterns and reduce the interpretability of the results. Conversely, more than 9 clusters tended to produce small fragmented groups with limited practical meaning. The feature space combined dynamic speed characteristics and spatial coordinates. After standardisation, spatial features were assigned a weight of 1.75, while dynamic features were assigned a weight of 1. This weighting favoured geographical continuity in the resulting clusters while still allowing differences in traffic behaviour to influence the clustering process.
The spawn points work as follows: each cluster is assigned a link that serves as its spawning point, from which the K-Means algorithm is initialized. These spawning points provide the initial centroids for the clustering process. During the first assignment step, all links in the network are assigned to the cluster whose centroid is the closest according to the selected distance metric.

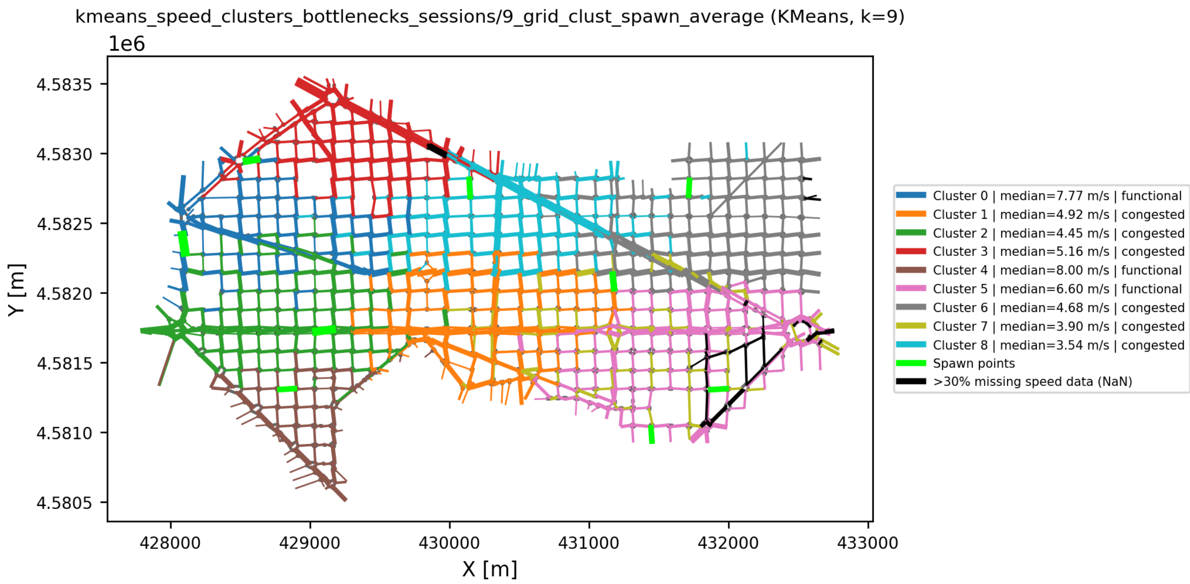
The centroids are then recomputed based on the assigned links, and the assignment-update process is repeated iteratively until convergence is reached. Convergence is achieved when the cluster assignments no longer change between successive iterations. By default, the Scikit-learn implementation of K-Means uses the k-means++ initialization strategy, which automatically selects well-distributed initial centroids to improve convergence and clustering quality. However, in this study, the spawn points are selected manually based solely on their spatial coordinates. This approach is preferred because it places the initial centroids in geographically meaningful locations, leading to a more interpretable and regionally coherent clustering of the network.

3.3.3 Results

For the sake of simplicity, we first analyse the results obtained by averaging over all available sessions. This provides a global view of the clustering behaviour and facilitates comparisons between methods.
To further investigate the influence of temporal variability, the analysis is then repeated on selected subsets of sessions. In particular, results are presented for all 101 sessions and for sessions 1–20 only, the latter providing a more homogeneous subset and allowing the stability of the identified clusters to be assessed under reduced variability.
Missing speed observations (NaN values) are handled using a tiered preprocessing strategy before clustering. For links with at most 30% missing observations, NaN values are replaced by the median speed of the corresponding link over time. Links with more than 30% missing observations are flagged as no-data links. Since the K-Means algorithm cannot operate on missing values, any remaining NaN entries are replaced by the global mean of the dataset before clustering. Consequently, these links are still assigned to clusters, although their temporal profiles become more neutral and therefore exert less influence on the clustering process. To indicate their lower data reliability, they are displayed in black in the network visualizations.



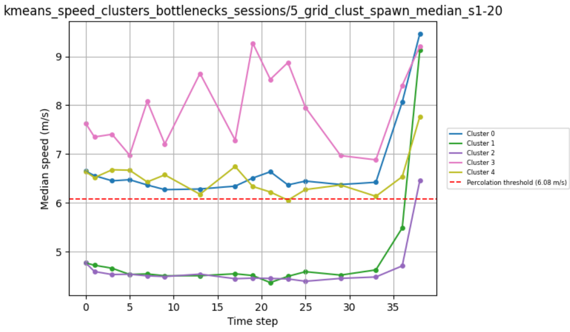
(a) $K = 5$



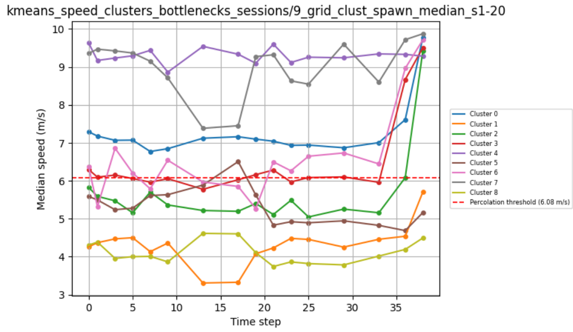
(b) $K = 9$

Figure 12: K-means clustering results for $K = 5$ and $K = 9$. Green links indicate the manually selected spawn points used for centroid initialization, while black links correspond to road segments with more than 30% missing speed observations.

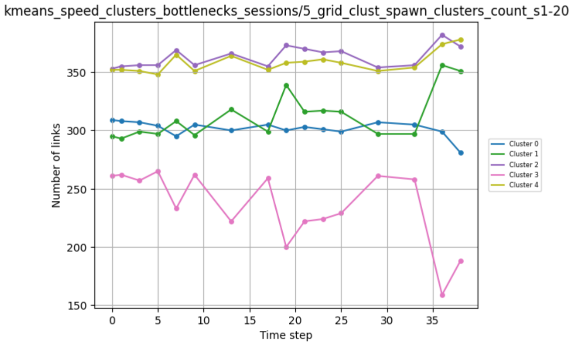
The followwing graphs show the comparative metrics used to analysze the different clustering configuration for the maps of Figure **??**.



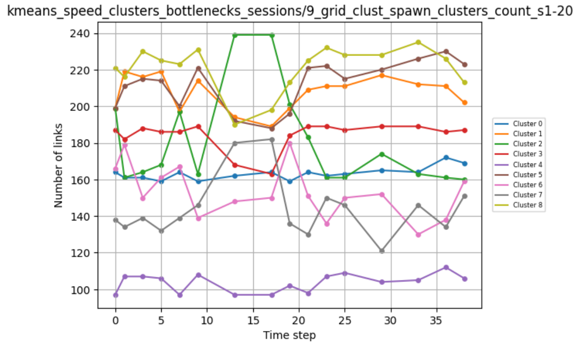
(a) Median speed, $K = 5$



(b) Median speed, $K = 9$



(c) Cluster size, $K = 5$



(d) Cluster size, $K = 9$

Figure 13: Evolution of the median speed and cluster sizes over time for the K-Means clustering with $K = 5$ and $K = 9$.

3.4 Approach 3— Grid K-means Clustering

3.4.1 Feature Construction and K-means Setup: Grid

K-means partitions the grid cells into K clusters that jointly encode *spatial position* and *average normalised speed*. The feature vector for each non-empty cell c is:

$$\mathbf{x}_c \; = \; \mathrm{StandardScaler}\Big(\big[\,\mathrm{cell}_x^{(c)},\; \mathrm{cell}_y^{(c)},\; \bar{r}_c\,\big]^\top\Big), \tag{9}$$

where $\mathrm{cell}_x^{(c)}$ and $\mathrm{cell}_y^{(c)}$ are the integer grid coordinates and $\bar{r}_c$ is the time-and-session-averaged normalised speed of cell c. Standard scaling normalises the three features to zero mean and unit variance so that spatial extent and speed range contribute equally to the Euclidean distance used by K-means.

K-means is fitted with k-means++ initialisation and 10 independent random restarts; the lowest-inertia solution is retained. Each resulting cluster is labelled *congested* if its mean normalised speed $\bar{r}_k < q_c = 0.52$, and *functional* otherwise. Congested clusters are shown with a hatched fill on maps.

The five cluster centroids in cell space are additionally mapped back to the nearest road segment on the simplified network (the *spawn points*), making the partition interpretable at segment level and providing a spatially well-distributed initialisation for further analyses.

3.4.2 Cluster Count: $K = 5$ vs. $K = 9$

Two values of K are evaluated on the same grid and feature set:

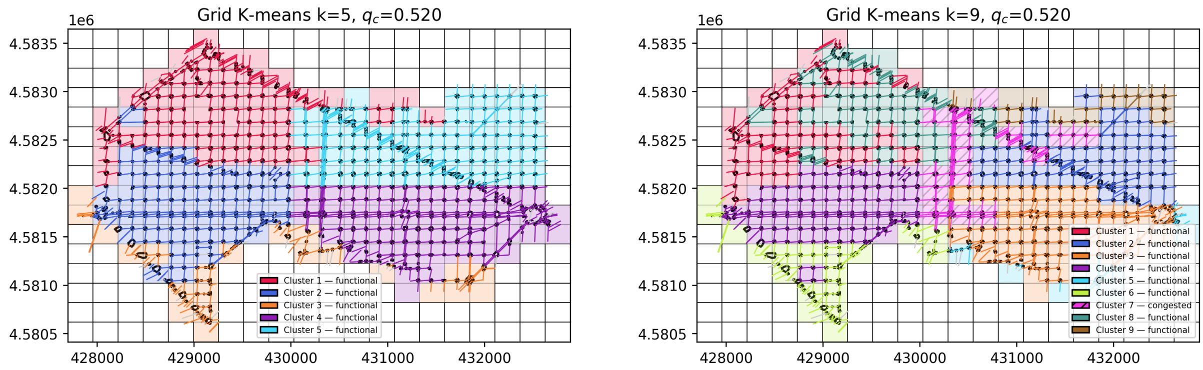
- $K = 9$: finer spatial granularity, better able to resolve distinct sub-congested zones within one geographic area. Risk: a single physically coherent congested zone may be split into several clusters with similar speed profiles, reducing interpretability.

- $K = 5$: broader zones aligned with the network's macro-scale structure (roughly one cluster per urban district), yielding five clearly distinct, stable partitions.

The preferred K is selected by comparing the three evaluation metrics and the visual coherence of the partition; the result is reported in Section **??**.

3.4.3 Results

Figure **??** shows the Grid K-means partition for $K = 5$ and $K = 9$ under $q_c = 0.52$.



(a) $K = 5$ (b) $K = 9$

Figure 14: Grid K-means partition at $q_c = 0.52$ (time-and-session-averaged normalised speed). Each colour represents one cluster. Clusters are labelled congested (hatched) if their mean $\bar{r}_k < 0.52$; since all time-averaged cluster speeds exceed 0.52, no hatching appears — the threshold captures only acute per-timestep drops below free-flow.

Figure **??** shows the median raw speed of each cluster tracked over the 15 representative timesteps for the preferred K.

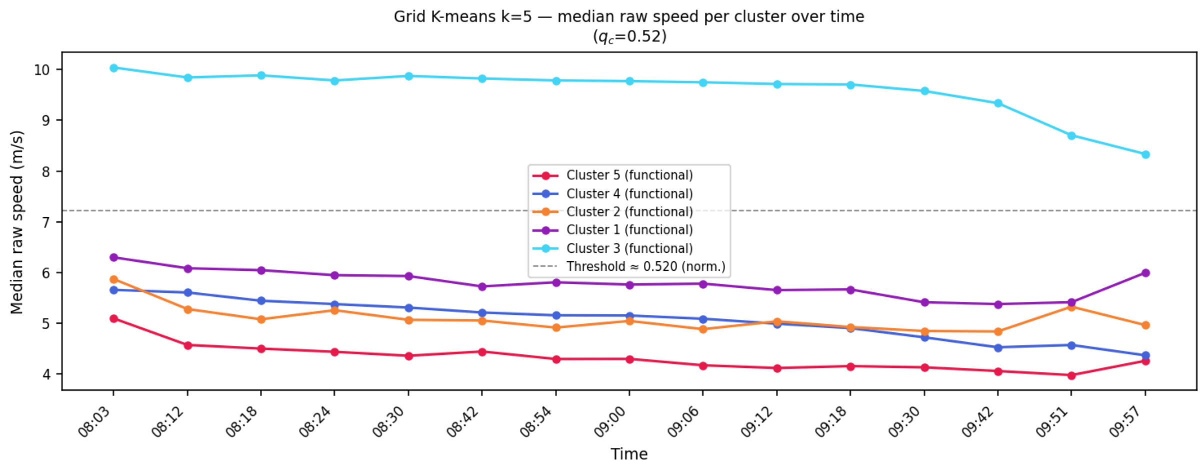


Figure 15: Grid K-means ($K = 5$, $q_c = 0.52$): median raw speed (m/s) per cluster over the 15 representative timesteps. Clusters are ordered by mean speed (ascending), so Cluster 1 is the slowest zone. All clusters remain above the percolation threshold ($\approx 4.5\,\mathrm{m/s}$ in raw speed) on a time-averaged basis, reflecting the moderate level of congestion in the Simbarca simulation.

Table ?? reports the three evaluation metrics for Approach 2 at the 15 representative timesteps using the preferred K. Clusters are ordered by mean speed (ascending), so cluster 1 is the most congested zone.

Table 2: Approach 2 metrics (Grid K-means, preferred K, $q_c = 0.52$). N_c: number of clusters labelled congested. S_k: size (grid cells) and $\tilde{v}_k$: median raw speed (m/s) of cluster k, ordered by ascending mean speed.

Time	N_c	S_1	S_2	S_3	S_4	S_5	$\tilde{v}_1$	$\tilde{v}_2$	$\tilde{v}_3$	$\tilde{v}_4$	$\tilde{v}_5$
08:03	0	42	41	37	46	22	5.10	5.66	5.87	6.30	10.04
08:12	0	42	41	37	46	22	4.57	5.61	5.28	6.09	9.84
08:18	0	42	41	37	46	22	4.50	5.44	5.08	6.05	9.89
08:24	0	42	41	37	46	22	4.44	5.38	5.26	5.95	9.79
08:30	0	42	41	37	46	22	4.36	5.31	5.07	5.93	9.87
08:42	0	42	41	37	46	22	4.44	5.21	5.06	5.73	9.82
08:54	0	42	41	37	22	46	4.30	5.16	4.92	9.79	5.81
09:00	0	42	41	37	46	22	4.30	5.15	5.05	5.76	9.77
09:06	0	42	41	37	46	22	4.17	5.09	4.89	5.78	9.75
09:12	0	42	41	37	46	22	4.12	4.99	5.04	5.65	9.71
09:18	0	42	41	37	46	22	4.16	4.91	4.93	5.67	9.71
09:30	0	42	41	37	46	22	4.13	4.72	4.85	5.41	9.58
09:42	0	42	41	37	46	22	4.06	4.53	4.84	5.38	9.34
09:51	0	42	41	37	46	22	3.98	4.57	5.33	5.42	8.70
09:57	1	42	41	37	46	22	4.26	4.37	4.97	6.00	8.34

3.5 Comparative Analysis

Figure ?? shows the two approaches side by side on a single map at a representative mid-rush timestep. The left panel is the fraction-based grid classification ($q_c = 0.52$); the right panel is the percolation network map computed on the simplified graph ($q_c = 0.525$, auto-computed from the full percolation sweep).

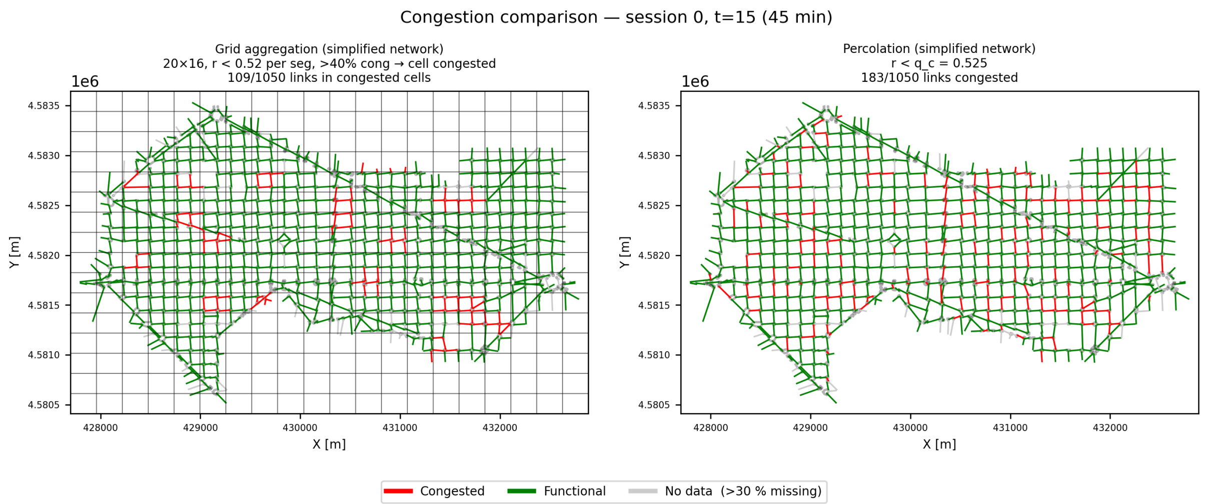


Figure 16: Side-by-side comparison of the two congestion-detection methods at timestep 15 (approximately 08:48). **Left**: fraction-based grid classification (20×16 grid, $q_c = 0.52$; cells coloured red if $> 40\,\%$ of segments have $r < 0.52$, green otherwise). **Right**: percolation network map on the simplified graph at $q_c = 0.525$ (auto-computed); segments coloured by SCC membership or congestion state.

Table ?? places the key metrics of both approaches side by side at seven representative timesteps spanning the full rush-hour window. The two methods are compared on the number of congested clusters they identify, the size of the largest congested zone, and the median speed within it. Qualitative differences in spatial layout are discussed with reference to Figures ?? and ??.

Table 3: Side-by-side comparison of Approach 1 (BFS, $q_c = 0.52$) and Approach 2 (Grid K-means, preferred K, $q_c = 0.52$) at seven representative timesteps. N: number of congested clusters. S_1: size (grid cells) of the largest congested cluster. $\tilde{v}_1$: its median raw speed (m/s).

	N congested clusters		S_1 (grid cells)		$\tilde{v}_1$ (m/s)	
Time	BFS	K-means	BFS	K-means	BFS	K-means
08:03	0	0	–	42	–	5.10
08:18	3	0	2	42	3.31	4.50
08:42	5	0	2	42	3.16	4.44
09:00	4	0	2	42	2.99	4.30
09:18	7	0	2	42	3.51	4.16
09:42	8	0	6	42	4.08	4.06
09:57	10	1	41	42	3.87	4.26

4 Evaluation

4.1 Evaluation framework

The three approaches are evaluated according to their ability to identify meaningful congested and uncongested regions within the road network. The comparison focuses on four complementary aspects: traffic-state homogeneity, spatial coherence, temporal consistency, and interpretability.

Quantitative metrics are introduced in Section ?? and are computed for a set of representative timesteps spanning the morning peak period. These metrics are used to assess how well each method captures the spatial organisation and temporal evolution of congestion.

The evaluation is performed on the subset of timesteps

$$\mathcal{T} = \{0, 3, 5, 7, 9, 13, 17, 19, 21, 23, 25, 29, 33, 36, 38\},$$

which covers the entire simulation window while providing finer temporal resolution during the congestion build-up phase.

In addition to the quantitative analysis, visual interpretability is considered. A desirable partition should generate spatially coherent regions that can be associated with major traffic corridors, recurrent bottlenecks, or urban districts.

Together, these elements provide a common framework for comparing the three approaches and identifying their respective strengths and limitations.

4.2 Metrics

The three approaches are evaluated using a common set of metrics designed to assess complementary aspects of the identified regions. Traffic-state metrics measure the homogeneity and separation of traffic conditions within clusters, spatial metrics characterise the extent and fragmentation of congestion across the network, and temporal metrics evaluate how these regions evolve throughout the morning peak period.

4.2.1 Traffic-State Metrics

The primary objective of the three approaches is to identify regions characterised by distinct traffic conditions. For each cluster k at timestep t, the traffic state is evaluated through the median speed

$$\tilde{v}_k(t) = \text{median}\left\{v_{ij}(t) : ij \in C_k(t)\right\}, \tag{10}$$

where $C_k(t)$ denotes the set of links (or grid cells) belonging to cluster k. The median is preferred to the mean because it is less sensitive to outliers and short-lived speed fluctuations.

The separation between the speed distributions of different clusters provides an indication of the ability of a method to distinguish congested and uncongested traffic states. Clusters exhibiting relatively homogeneous median speeds are considered more representative of coherent traffic conditions.

4.2.2 Spatial Metrics

The spatial organisation of congestion is evaluated through the size and distribution of the identified regions. The size of cluster k is defined as

$$S_k(t) = |C_k(t)|. \tag{11}$$

For the BFS-based approach, cluster size corresponds to the number of congested grid cells belonging to the same connected component. For the K-means approaches, it corresponds to the number of links or grid cells assigned to a given cluster.

In addition, the number of congested regions $N_c(t)$ is used to characterise the spatial fragmentation of congestion. Large values of $N_c(t)$ indicate multiple independent congestion zones, whereas small values correspond to more concentrated congestion patterns.

4.2.3 Temporal Metrics

Congestion is inherently dynamic, making the temporal evolution of the identified regions an important aspect of the evaluation. Rather than introducing new indicators, the temporal analysis relies on the evolution of the traffic-state and spatial metrics defined above.

The metrics are therefore computed for a subset of representative timesteps covering the entire morning peak period. The evolution of cluster median speeds $\tilde{v}_k(t)$ provides insight into the formation and dissipation of congestion, while changes in cluster sizes $S_k(t)$ reveal the spatial growth, contraction, merging, or fragmentation of congested regions over time.

Temporal metrics should thus be interpreted as the dynamic counterpart of the traffic-state and spatial metrics. A method is considered temporally consistent if it captures these transitions without producing abrupt or implausible changes in the spatial partition of the network.

4.3 Comparison of the Three Approaches

4.3.1 Traffic-State Homogeneity

4.3.2 Spatial Coherence

4.3.3 Temporal Behaviour

4.3.4 Interpretability

4.4 Limitations

4.4.1 Missing Data

4.4.2 Demand and Session Structure

4.4.3 Parameter Sensitivity

4.4.4 Methodological Limitations

5 Conclusion

5.1 Summary of findings

5.2 Future work

A Implementation overview

B Parameter reference